# Milestone 1: Formula 1 Podium Prediction Companion

## Analytical report - leakage-safe prediction after qualifying and before race start

The immutable 1950-2024 snapshot contains 26,759 raw result rows. A deterministic shared-drive rule resolves 85 duplicate driver-race groups, leaving 26,668 unique predictions.

Primary metric: PR-AUC, because podiums are a minority class. ROC-AUC is secondary; F1, precision, and recall use validation-selected thresholds.

# Audit and cleaning

Literal \N values are parsed as missing. Dates and qualifying durations receive semantic types. All audited primary keys are unique and non-null, and all audited foreign keys reconcile. Raw CSV files are never mutated. For shared drives we prefer a row with grid > 0, then best final classification, then lowest resultId.

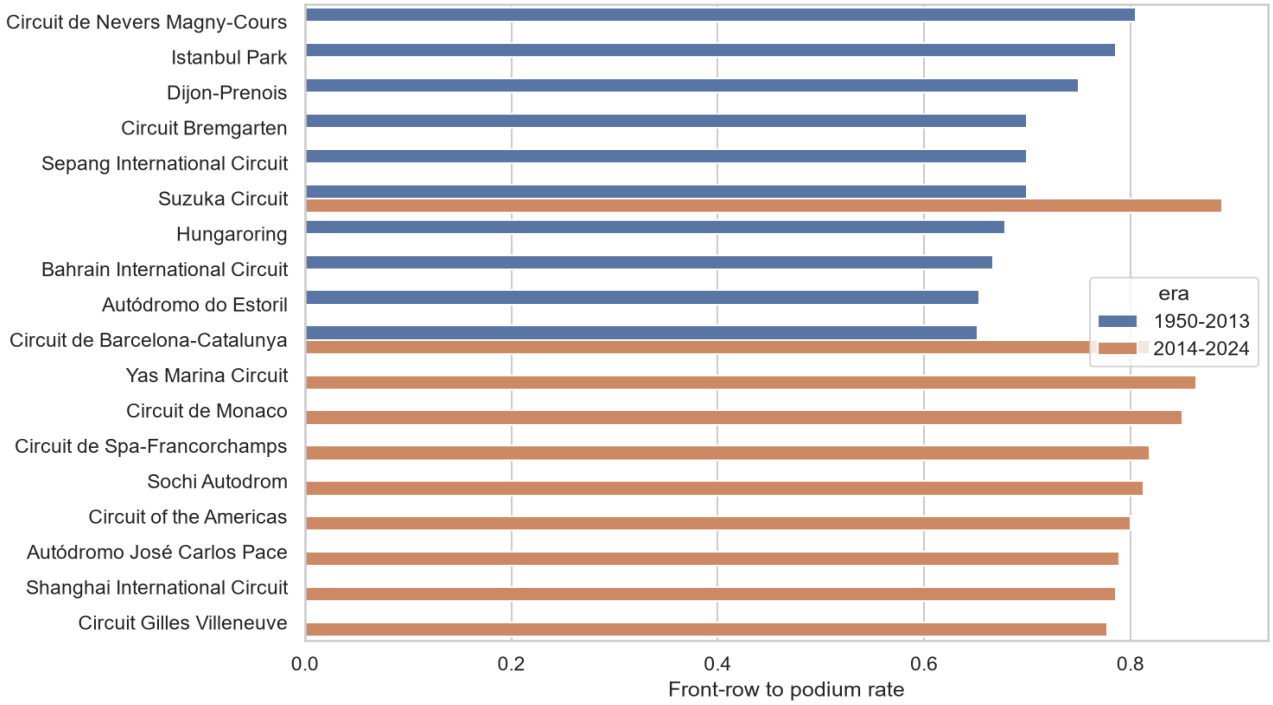



Qualifying coverage drifts strongly across history. The model therefore keeps an explicit missingness indicator, uses fitted imputation, and treats grid as the most consistent starting-order feature.

# Data-engineering answers

## 1. Front-row conversion by circuit

A front-row start is defined by actual grid positions 1-2, not qualifying position, because penalties can change the grid. Results are joined through races to circuits and reported separately for pre-2014 and 2014-2024, with at least ten starts per circuit-era.

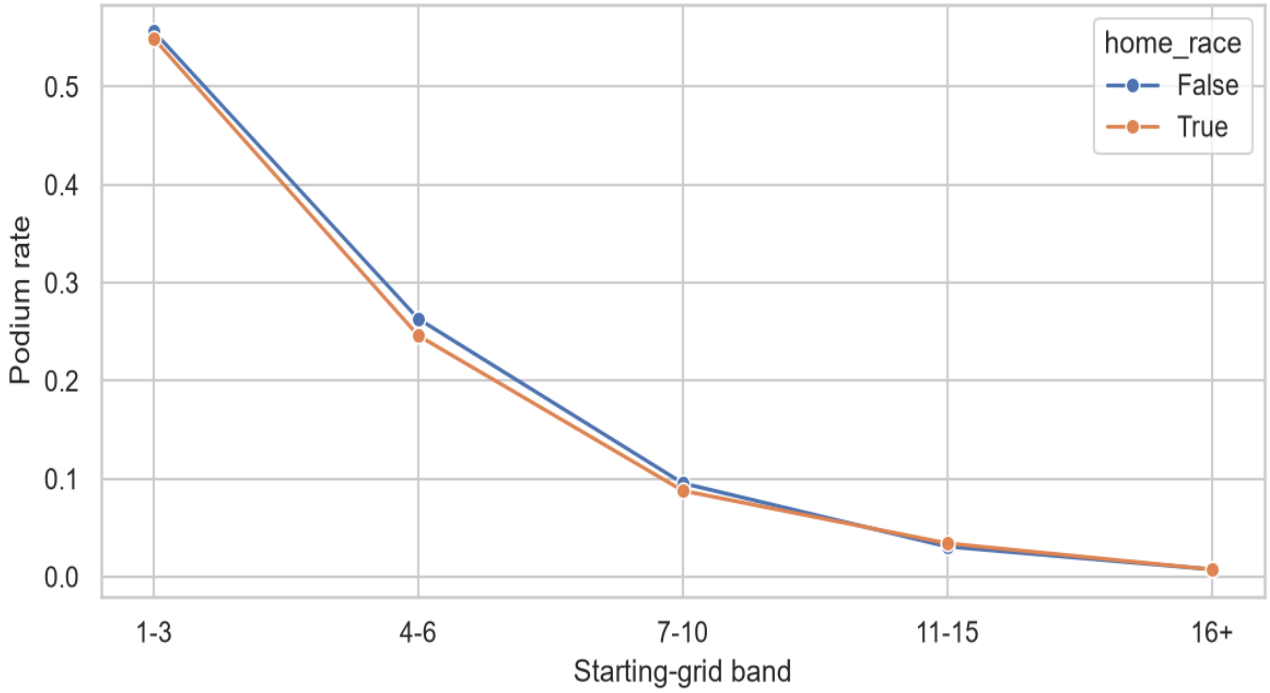



| era | circuit_name | mean | count |
|---|---|---|---|
| 1950-2013 | Circuit de Nevers Magny-Cours | 0.806 | 36.000 |
| 1950-2013 | Istanbul Park | 0.786 | 14.000 |
| 1950-2013 | Dijon-Prenois | 0.750 | 12.000 |
| 1950-2013 | Circuit Bremgarten | 0.700 | 10.000 |
| 1950-2013 | Sepang International Circuit | 0.700 | 30.000 |
| 2014-2024 | Suzuka Circuit | 0.889 | 18.000 |
| 2014-2024 | Yas Marina Circuit | 0.864 | 22.000 |
| 2014-2024 | Circuit de Monaco | 0.850 | 20.000 |
| 2014-2024 | Circuit de Barcelona-Catalunya | 0.818 | 22.000 |
| 2014-2024 | Circuit de Spa-Francorchamps | 0.818 | 22.000 |

Answer: the ranking changes across eras. Circuit de Nevers Magny-Cours leads 1950-2013 at 80.6%, while Suzuka Circuit leads 2014-2024 at 88.9% under the stated minimum-sample rule.

## 2. Home-race podium rate controlling for grid

Driver nationality is mapped to circuit country, then home and away podium rates are compared within the same starting-grid bands. This stratification controls the largest observed pre-race advantage without claiming a causal home effect.

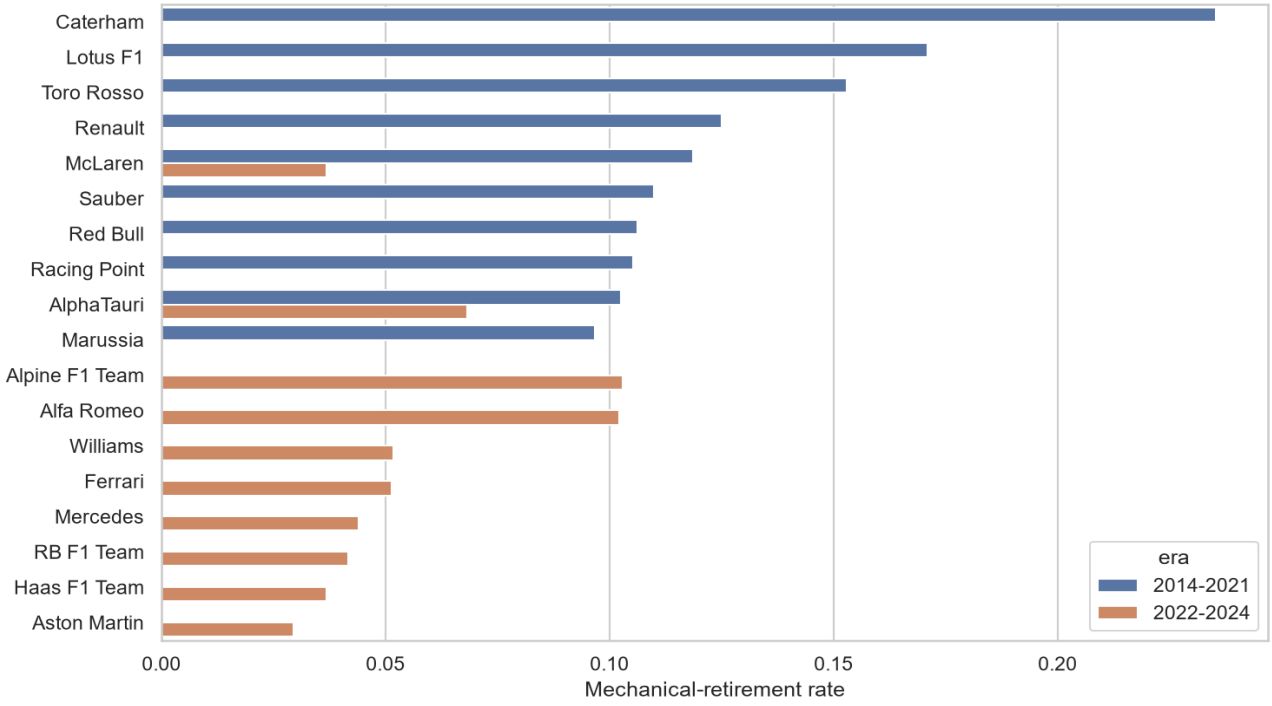



| grid_band | home_race | mean | count |
|---|---|---|---|
| 1-3 | False | 0.557 | 3060.000 |
| 1-3 | True | 0.549 | 264.000 |
| 4-6 | False | 0.263 | 3016.000 |
| 4-6 | True | 0.246 | 260.000 |
| 7-10 | False | 0.095 | 3990.000 |
| 7-10 | True | 0.088 | 387.000 |
| 11-15 | False | 0.031 | 4933.000 |
| 11-15 | True | 0.034 | 469.000 |
| 16+ | False | 0.008 | 6904.000 |
| 16+ | True | 0.007 | 936.000 |

Answer: after comparing like starting-grid bands, home drivers do not show a consistent podium advantage. Their rates are slightly lower in four bands and only marginally higher in grid positions 11-15; this descriptive result does not establish causality.

## 3. Mechanical retirements

Results are joined to status and constructor. Only documented component failures count; crashes, driver errors, disqualifications, and regulation decisions are excluded. Rates use race starts as the denominator and require at least 20 starts per constructor-era.




| era | constructor_name | mean | sum | count |
|---|---|---|---|---|
| 2014-2021 | Caterham | 0.235 | 8.000 | 34.000 |
| 2014-2021 | Lotus F1 | 0.171 | 13.000 | 76.000 |
| 2014-2021 | Toro Rosso | 0.153 | 37.000 | 242.000 |
| 2014-2021 | Renault | 0.125 | 25.000 | 200.000 |
| 2014-2021 | McLaren | 0.119 | 38.000 | 320.000 |
| 2014-2021 | Sauber | 0.110 | 22.000 | 200.000 |
| 2014-2021 | Red Bull | 0.106 | 34.000 | 320.000 |
| 2022-2024 | Alpine F1 Team | 0.103 | 14.000 | 136.000 |
| 2022-2024 | Alfa Romeo | 0.102 | 9.000 | 88.000 |
| 2022-2024 | AlphaTauri | 0.068 | 6.000 | 88.000 |
| 2022-2024 | Williams | 0.052 | 7.000 | 135.000 |
| 2022-2024 | Ferrari | 0.051 | 7.000 | 136.000 |
| 2022-2024 | Mercedes | 0.044 | 6.000 | 136.000 |
| 2022-2024 | RB F1 Team | 0.042 | 2.000 | 48.000 |

Answer: Caterham has the highest mechanical-retirement rate in 2014-2021 (23.5%); Alpine F1 Team leads 2022-2024 (10.3%). These rates use all starts, not only retirements, as the denominator.

# Features and leakage control

Inputs cover grid, qualifying, historical form, prior standings, and race context. Form is shifted before rolling, while standings after one race are attached only to the next race in that season. Current-race position, points, laps, status, pit stops, and fastest lap are excluded from the feature matrix. The chronological split is training through 2019,

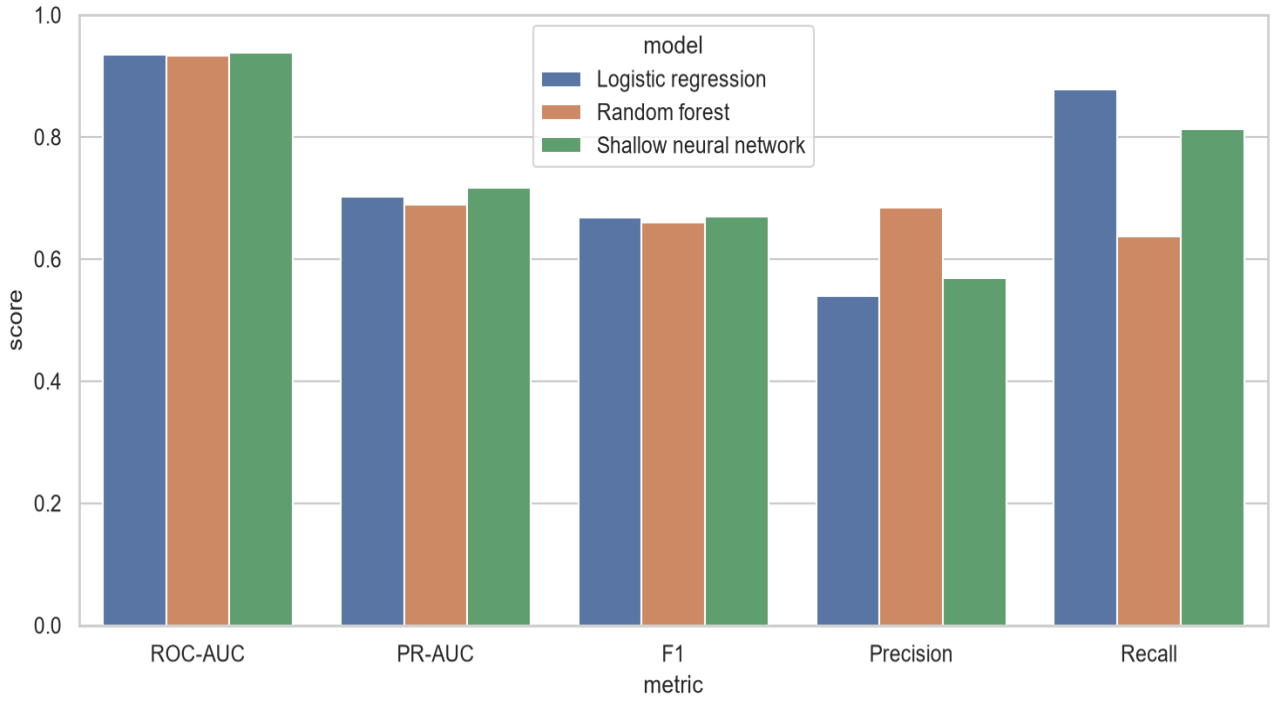
validation in 2020-2021, and unseen testing in 2022-2024.

## Preprocessing experiments

| preprocessing | validation_ROC-AUC | validation_PR-AUC |
|---|---|---|
| median + scaling | 0.926 | 0.728 |
| median, no scaling | 0.924 | 0.724 |
| zero + scaling | 0.927 | 0.725 |

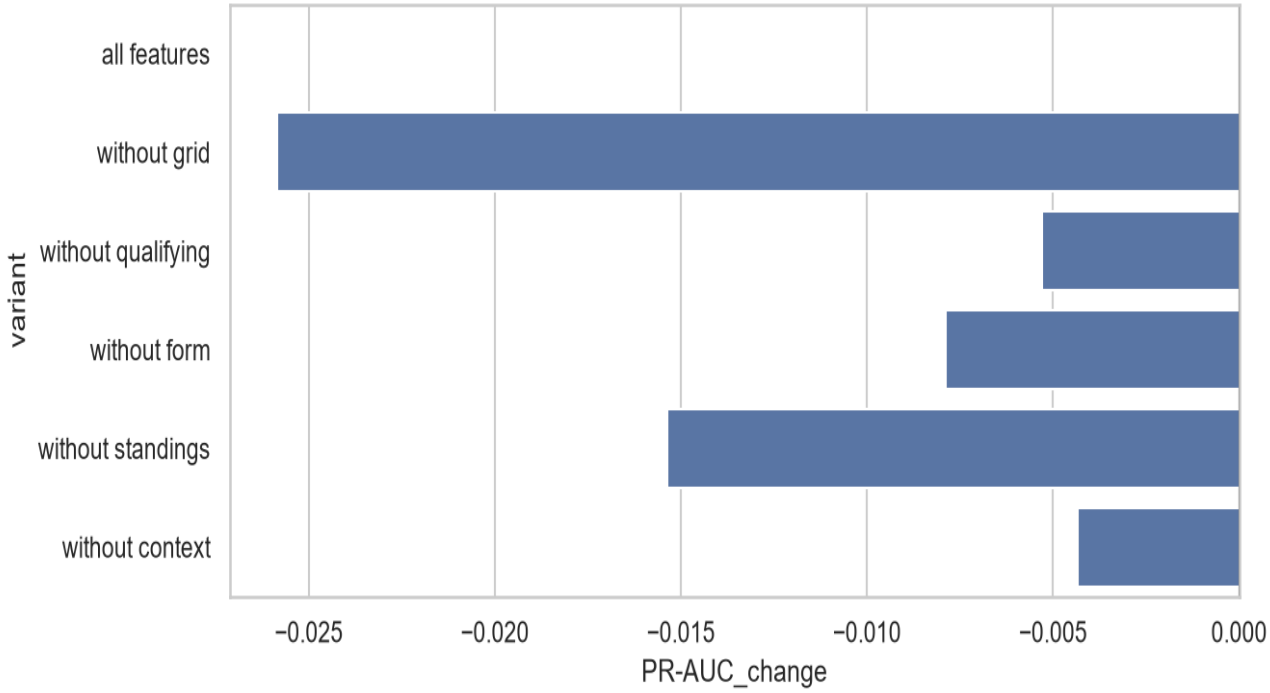
# Model comparison

Logistic regression is the baseline; random forest tests nonlinear interactions; and the shallow MLP has two hidden layers (32 and 12 units). All models use the same rows and feature set. Hyperparameters and thresholds are chosen without consulting the test labels.




| model | split | threshold | ROC-AUC | PR-AUC | F1 | Precision | Recall |
|---|---|---|---|---|---|---|---|
| Logistic regression | train | 0.655 | 0.906 | 0.561 | 0.580 | 0.468 | 0.763 |
| Logistic regression | validation | 0.655 | 0.926 | 0.728 | 0.703 | 0.647 | 0.769 |
| Logistic regression | test | 0.655 | 0.935 | 0.703 | 0.668 | 0.539 | 0.877 |
| Random forest | train | 0.640 | 0.988 | 0.919 | 0.829 | 0.827 | 0.831 |
| Random forest | validation | 0.640 | 0.924 | 0.702 | 0.709 | 0.757 | 0.667 |
| Random forest | test | 0.640 | 0.933 | 0.689 | 0.660 | 0.684 | 0.637 |
| Shallow neural network | train | 0.240 | 0.912 | 0.585 | 0.578 | 0.462 | 0.771 |
| Shallow neural network | validation | 0.240 | 0.928 | 0.738 | 0.713 | 0.677 | 0.752 |
| Shallow neural network | test | 0.240 | 0.938 | 0.716 | 0.669 | 0.568 | 0.814 |

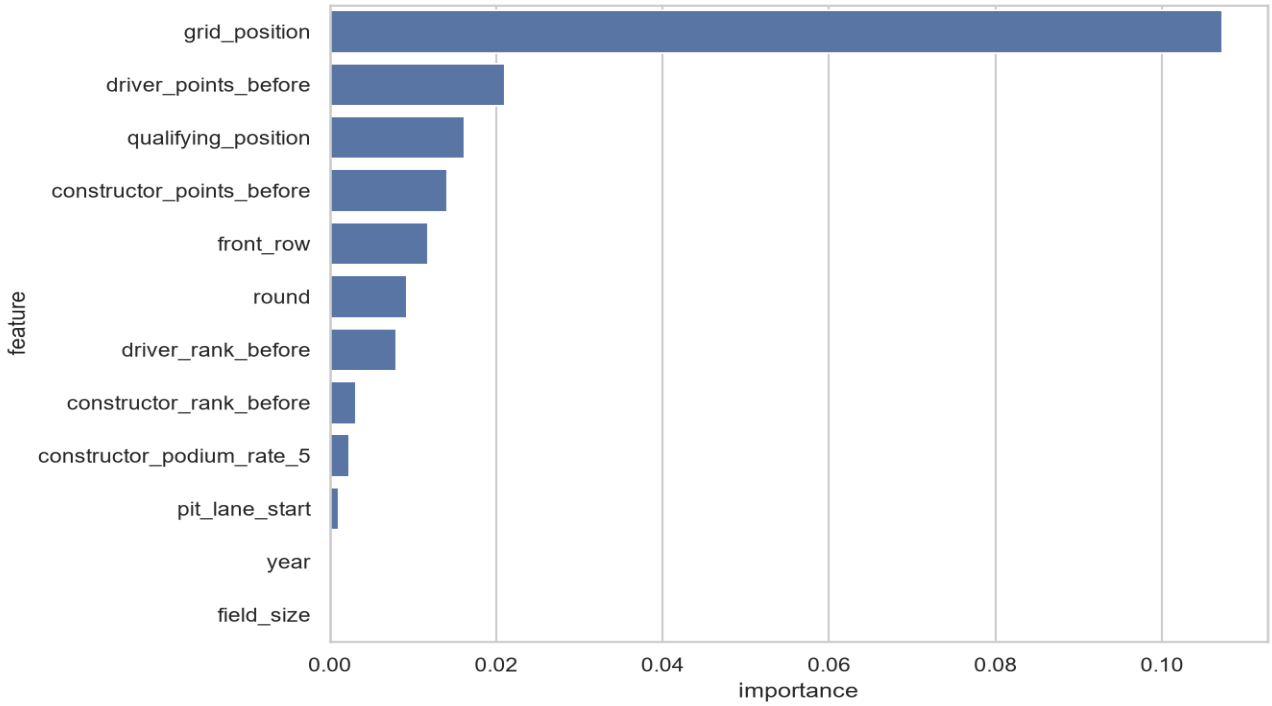
# Feature-group ablation

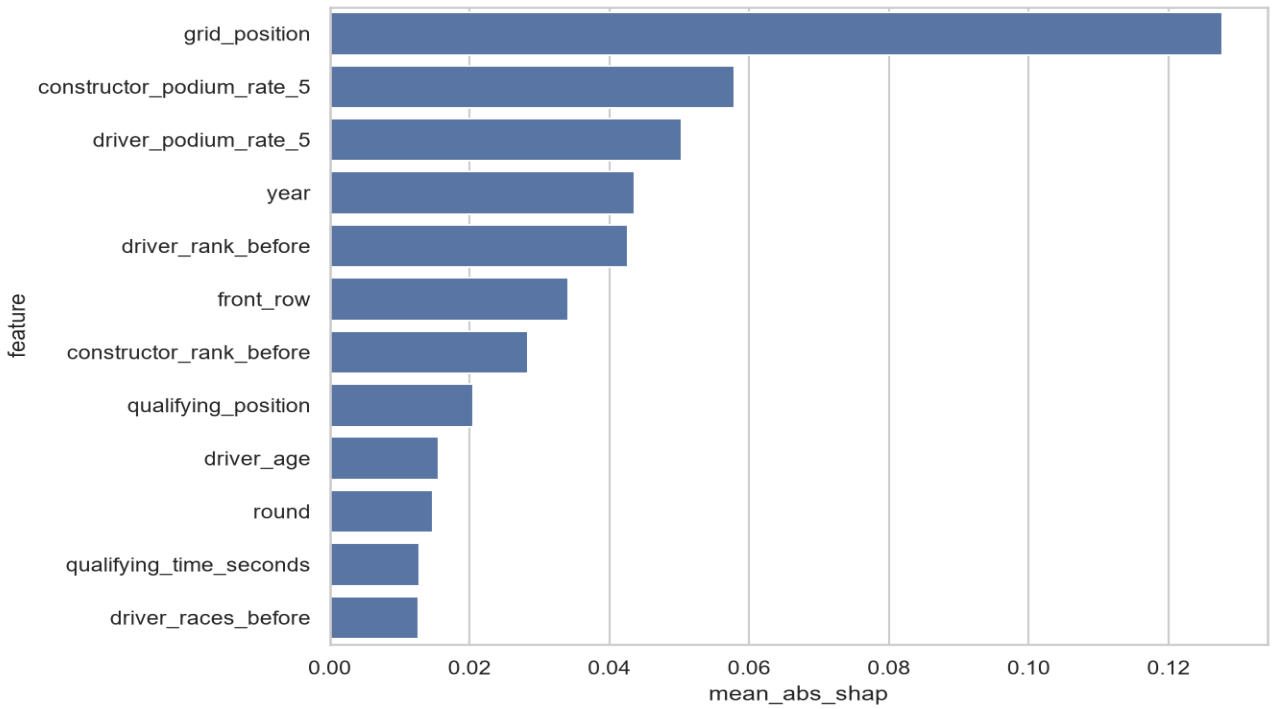


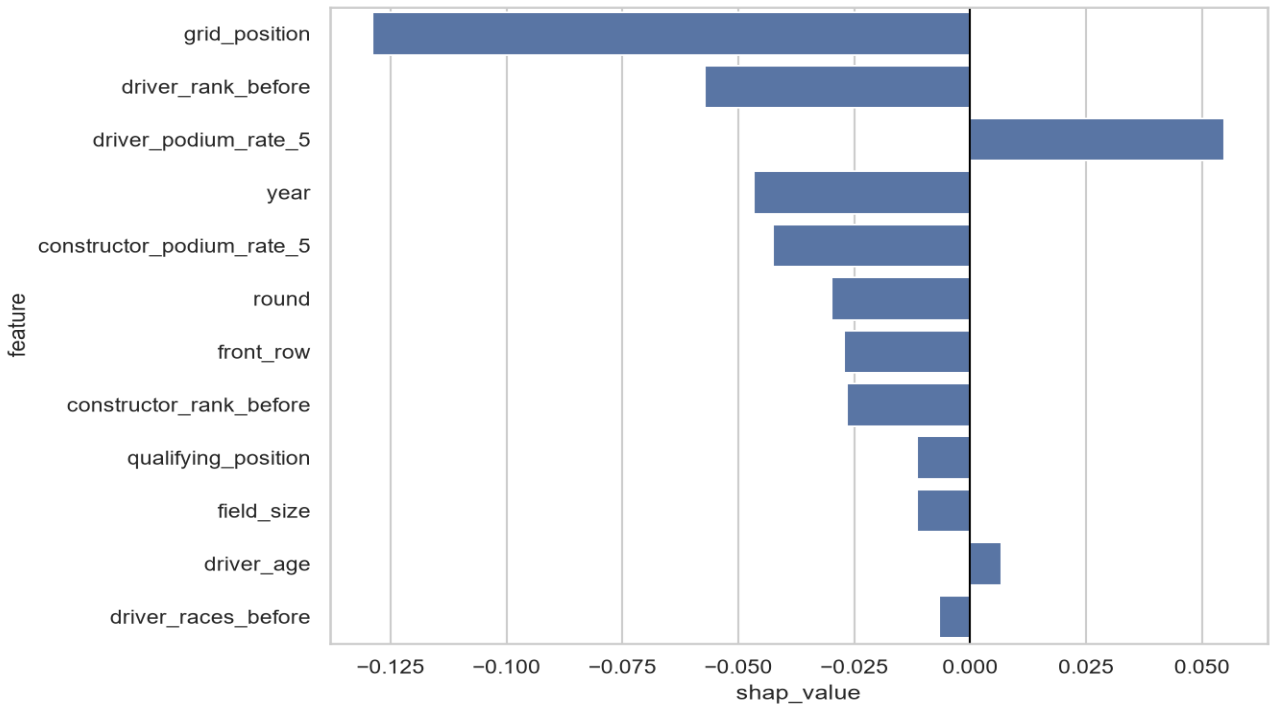

| variant | features | validation_PR-AUC | PR-AUC_change |
|---|---|---|---|
| all features | 17.000 | 0.728 | 0.000 |
| without grid | 13.000 | 0.702 | -0.026 |
| without qualifying | 14.000 | 0.722 | -0.005 |
| without form | 14.000 | 0.720 | -0.008 |
| without standings | 13.000 | 0.712 | -0.015 |
| without context | 14.000 | 0.723 | -0.004 |

# Explainability

Permutation importance and global SHAP describe model reliance across validation records; local SHAP explains one prediction. These are associations, not causal effects: importance does not prove that changing a feature causes a podium.






# Inference and limitations

The reusable inference function was exercised with complete and partially missing records. The selected validation model was Shallow neural network; missing fields are handled by its training-fitted imputer.

Limitations include historical qualifying-coverage drift, constructor identity resets after renames, incomplete nationality mapping, conservative classification of ambiguous failure statuses, and era sensitivity. Predictions are probabilistic associations, not guarantees.